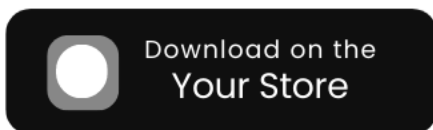
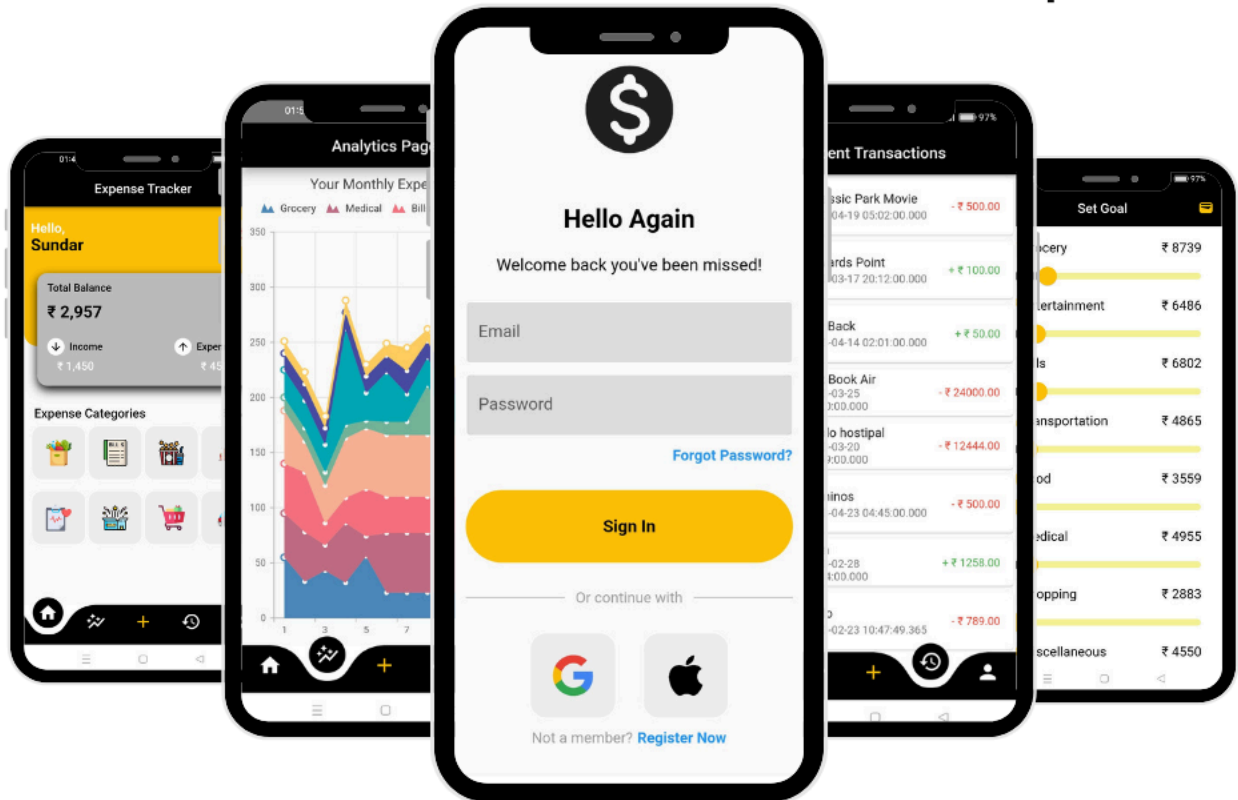




Now Add.
Analyze &
Manage
Expenses



THE EXPENSE TRACKER

Proudly Made by Amit Masal, Rhea Yadav and Naman Jain

Members of Group 14

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Naman Jain - 225501

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Project Overview

Here are the tools, techniques, packages and softwares we have used to make this app a reality.

- When we were designing the UI for our app, instead of making the UI using flutter we used **Figma** to design the pages so that we can get a clear view of going ahead with our project.
- Used **14 packages** to make this a possibility. Here is the list of the packages.
- Used **Microsoft Visual Studio Code** to develop our app.
- And use **GitHub** to store and maintain our app.

1. Introduction

Managing finances can be a troublesome task for many individuals. The solution to this problem is a finance management application. The app is designed to help you take control of your finances and reach your financial goals. With the easy-to-use interface, you can easily track your income and expenses. You can add transactions, set budgets, and view detailed reports to gain insights into your spending habits. The app is perfect for people of all financial backgrounds, whether you're a student, a working professional, or retired. With this app, managing finances will no longer be a stress, but rather an easy and efficient process.

2. Front End & Back End

The frontend and backend technologies that have been employed in our Mobile Application Development project for our **expense tracker** app are Firestore for the backend and Flutter for the frontend.

The Google-developed open-source Flutter framework is used to create fast, cross-platform mobile apps for iOS and Android. A single codebase can be used to create stunning and responsive user interfaces using Flutter, which simplifies application deployment and maintenance. Firestore is a NoSQL document-oriented database that is part of the Firebase suite of services, also developed by Google. Firestore is a cloud-based database that allows developers to store and sync data in real-time across multiple platforms and devices. Firestore provides a powerful API for querying and manipulating data, which makes it easy to build scalable and robust backend services for mobile applications.



In the context of an expense tracker app, the frontend built with Flutter can provide a clean and intuitive user interface for entering and managing expenses, while the backend built with Firestore can store and retrieve expense data securely in the cloud. Firestore can also provide features such as data synchronization, real-time updates, and data analysis, which can be used to build powerful features such as expense reports and analytics.

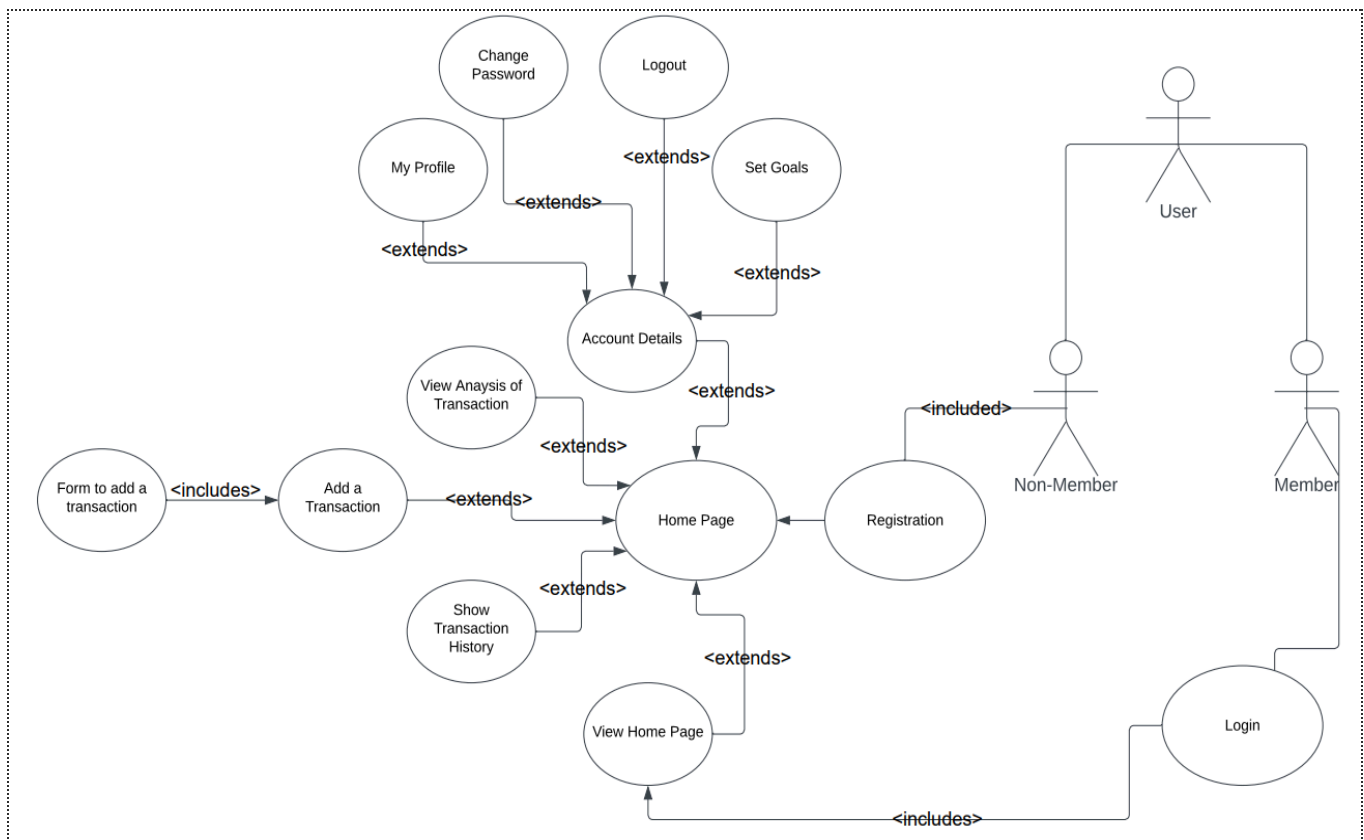
Overall, the combination of Flutter for the frontend and Firestore for the backend provides a powerful and efficient platform for building high-quality and scalable mobile applications.

3. Process Model

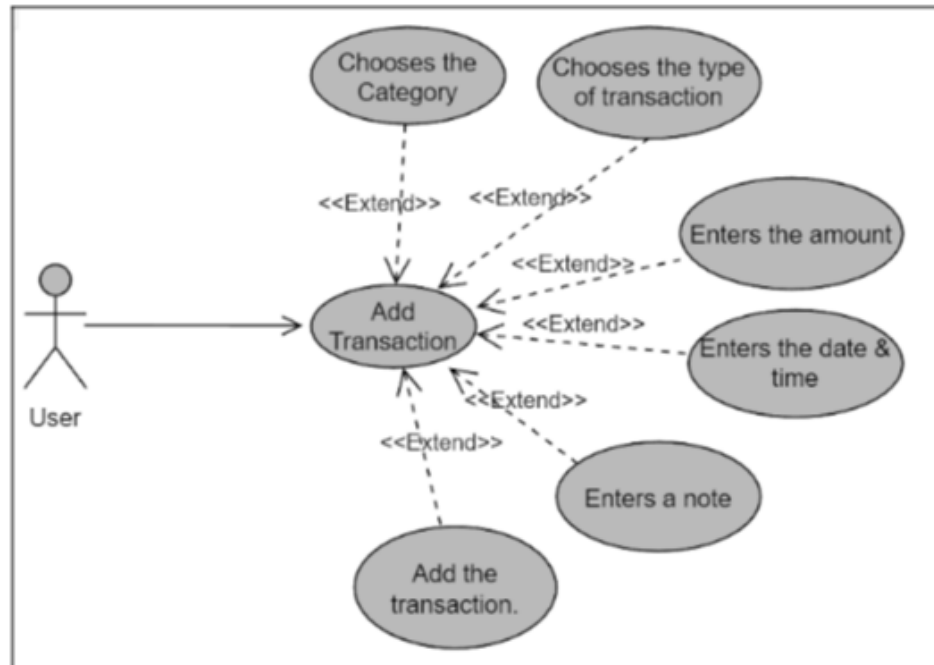
- We have used an Increment Model, also known as an iterative or agile development model, is a software development process where the project is divided into multiple iterations or increments.
- Each increment focuses on delivering a small set of features or functionality, and each iteration builds upon the previous one.
- The increment model allows for more flexibility and adaptability in the development process, as changes and adjustments can be made during each iteration based on feedback and testing.
- In our expense tracker project, we used an increment model to gradually develop and add features to the app in a systematic and efficient manner.
- Each increment focused on delivering a specific set of features or improvements, and each iteration allowed us to refine and improve the app based on feedback from users and testing.
- Overall, using an increment model allowed us to develop a high-quality and user-friendly app that meets the needs of our users while staying within our time and resource constraints.

4. Diagrams

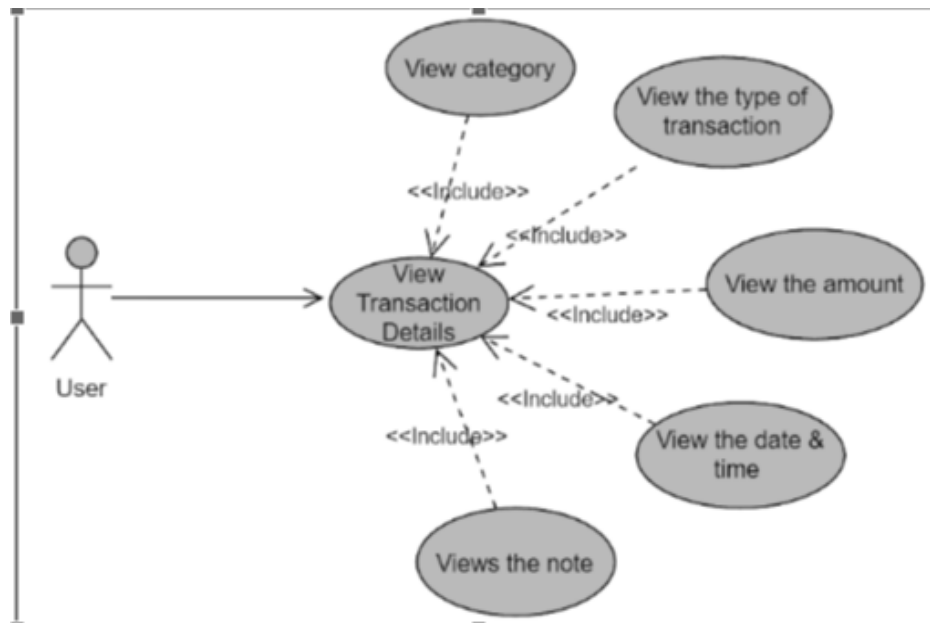
Use-case Diagram for complete app.



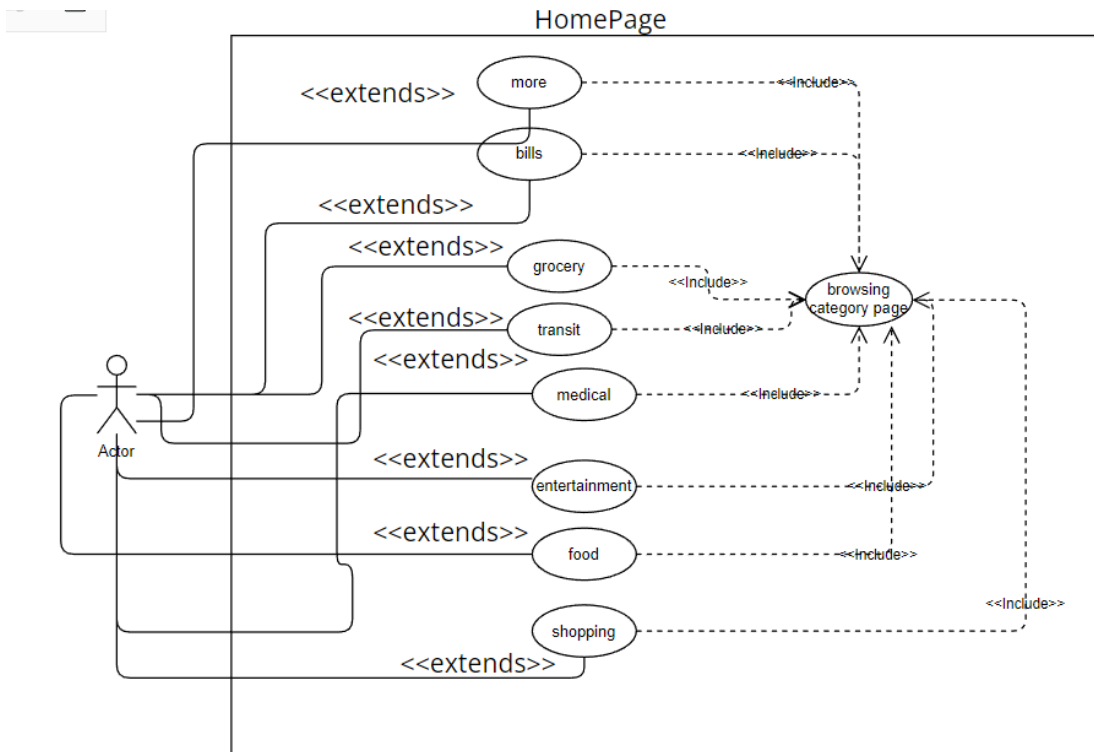
1. Insert Transaction Page



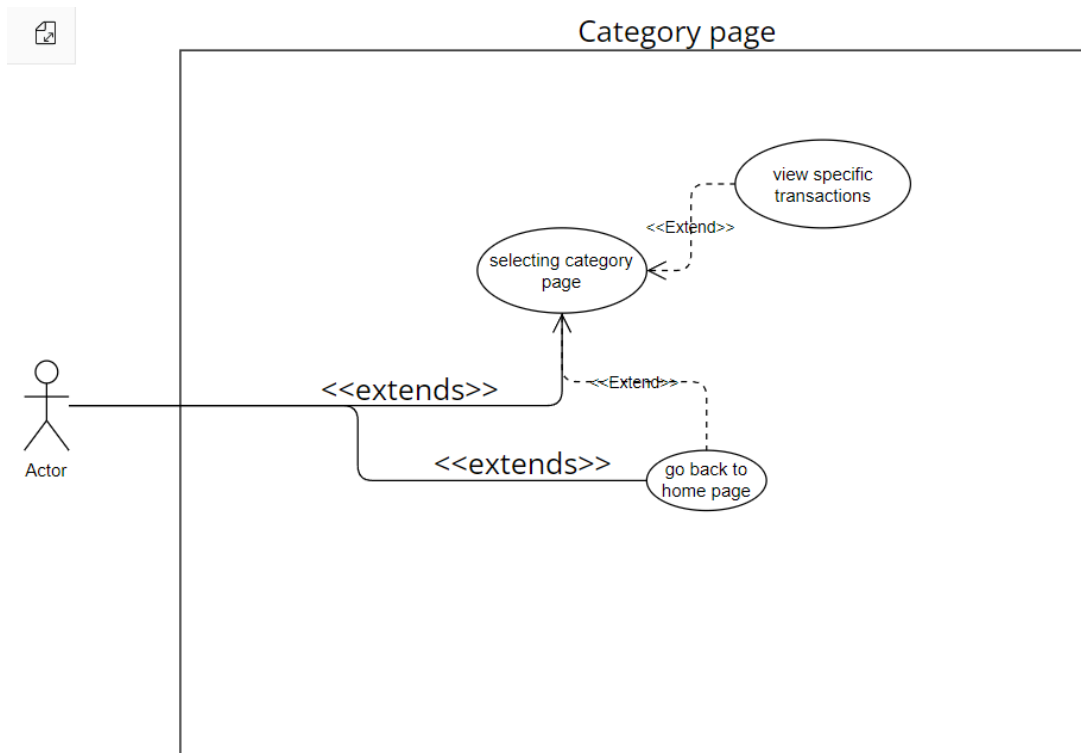
2. View Transaction Page



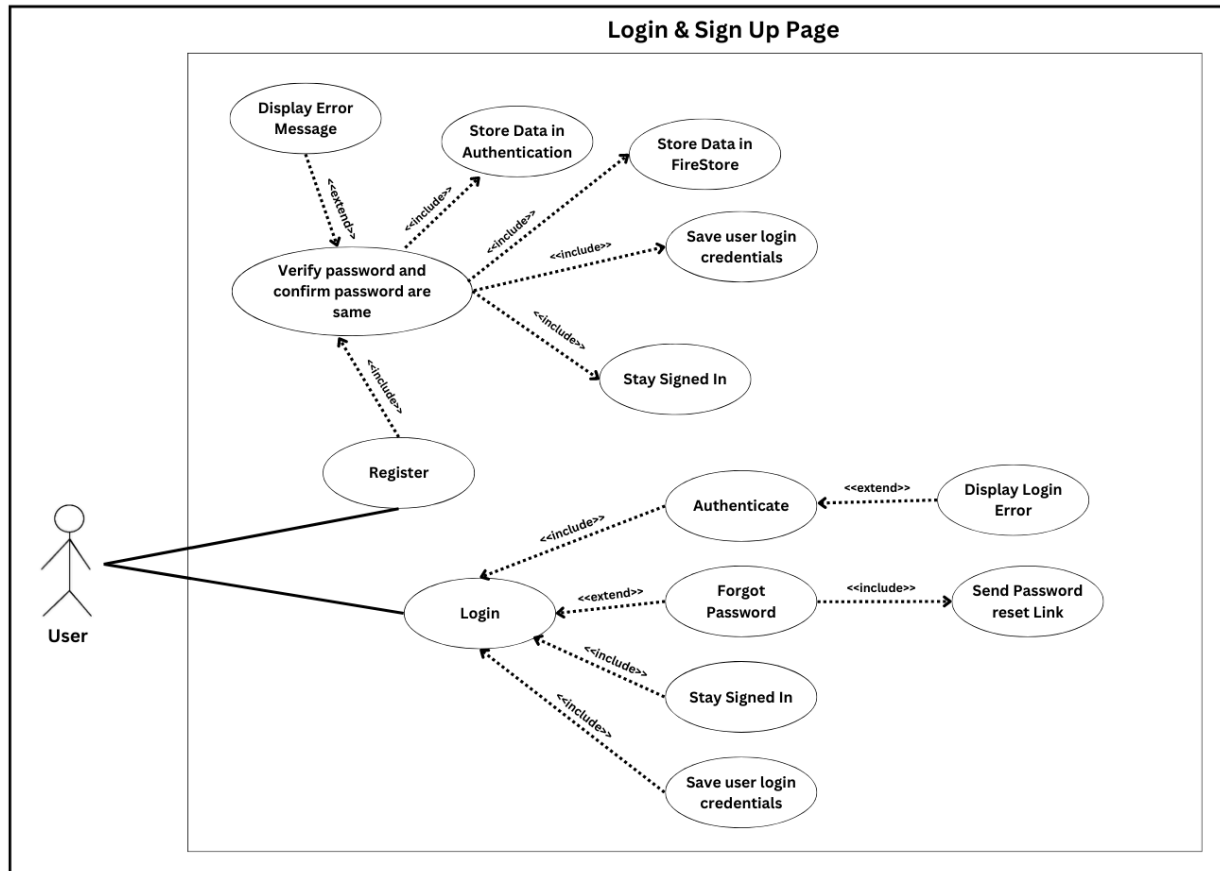
3. Home Page



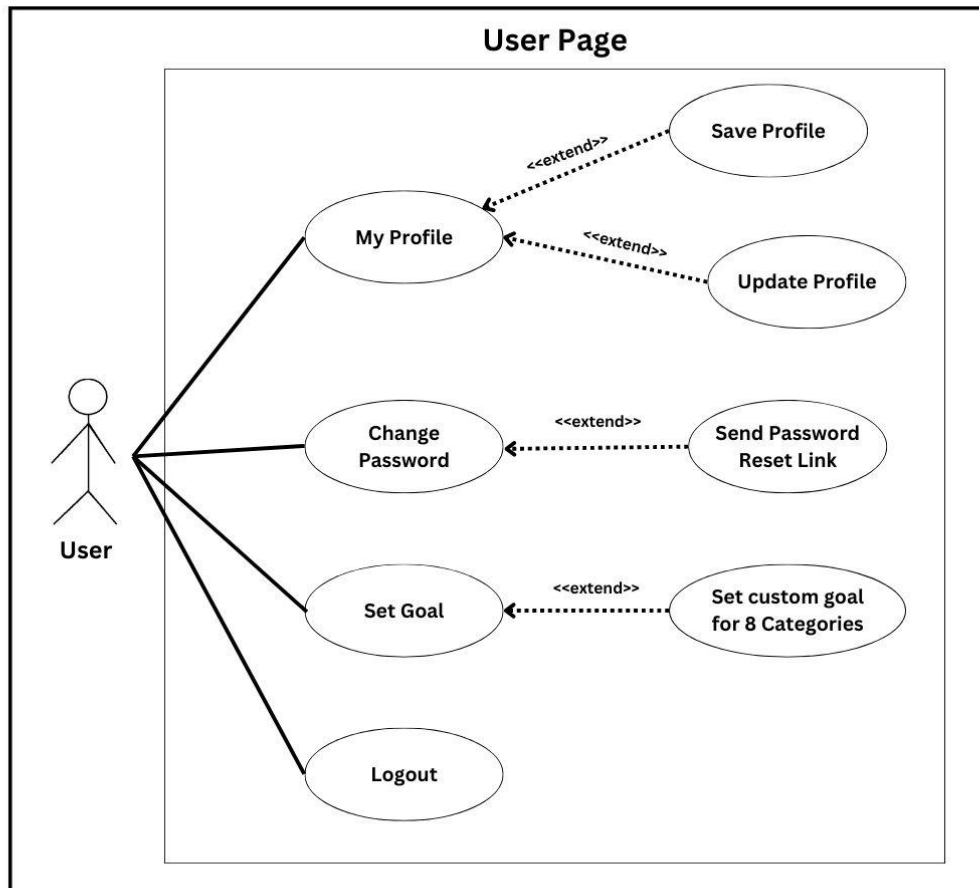
4. Category Page



5. Login & Sign Up

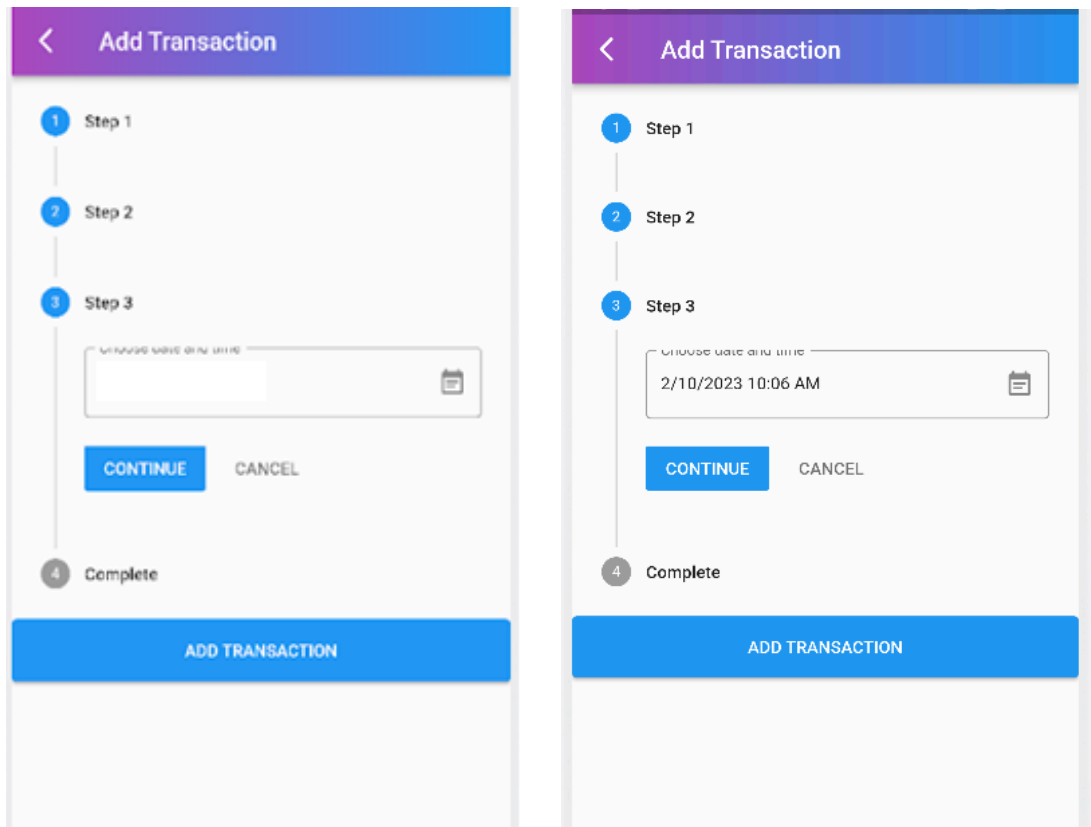


6. User Page



5. Errors & Debug

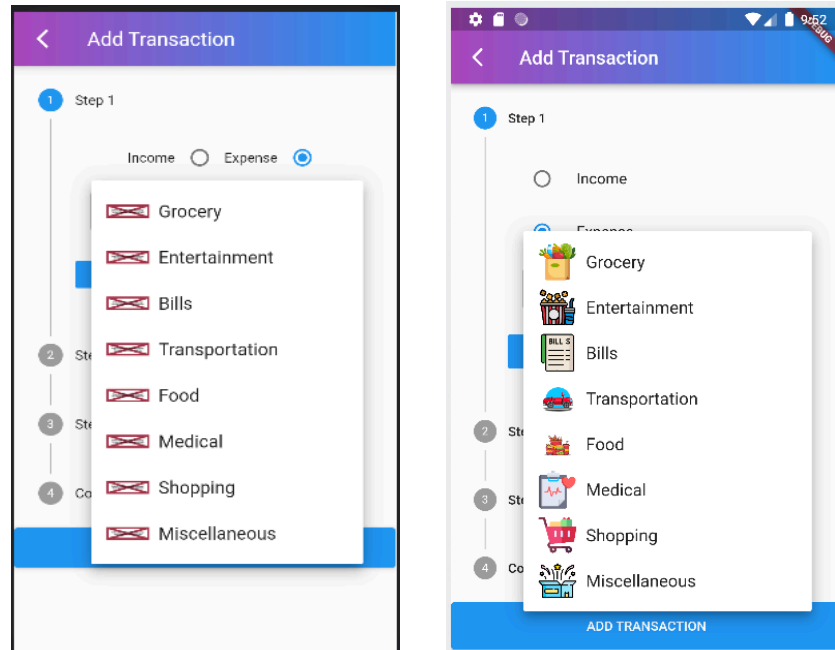
1. Add Transaction Page(1)



Error: Selected Date & time not displayed in the textbox

Solution: Updated the latest version of the package.

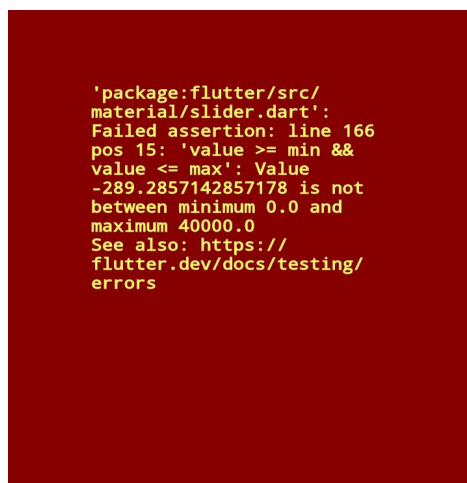
2. Add Transaction Page(2)



Error: Images not being displayed

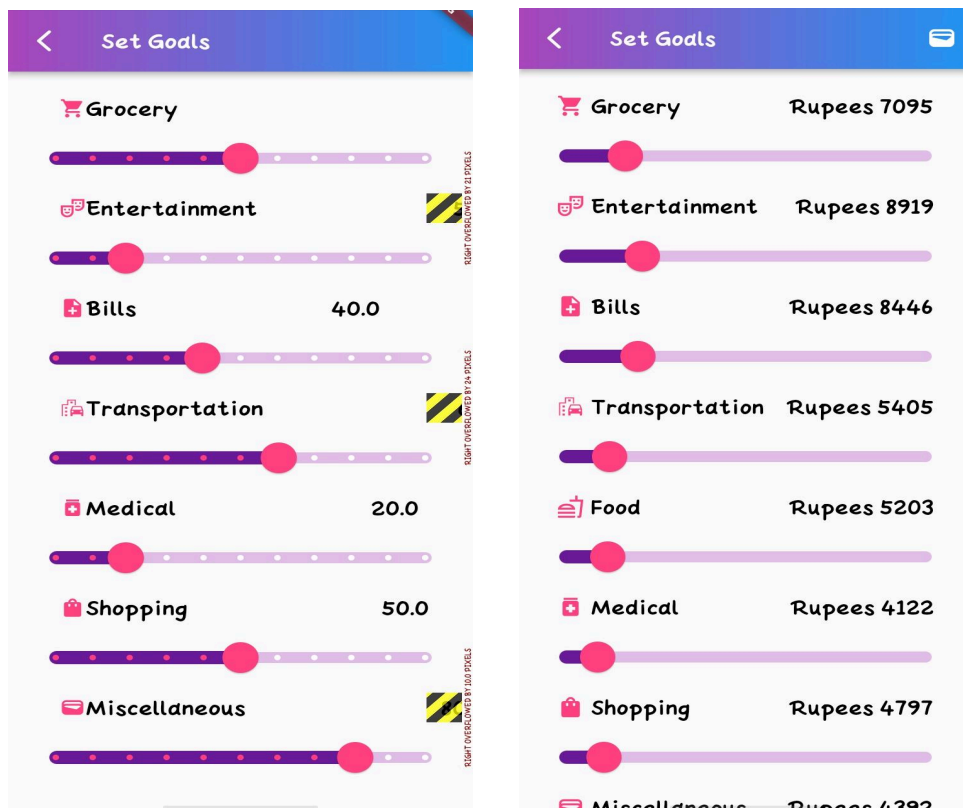
Solution: Made a separate folder for the assets and added its path to pubspec.yaml

3. Set Goals Page (1)



Solution: Min and Max of each slider was adjusted again.

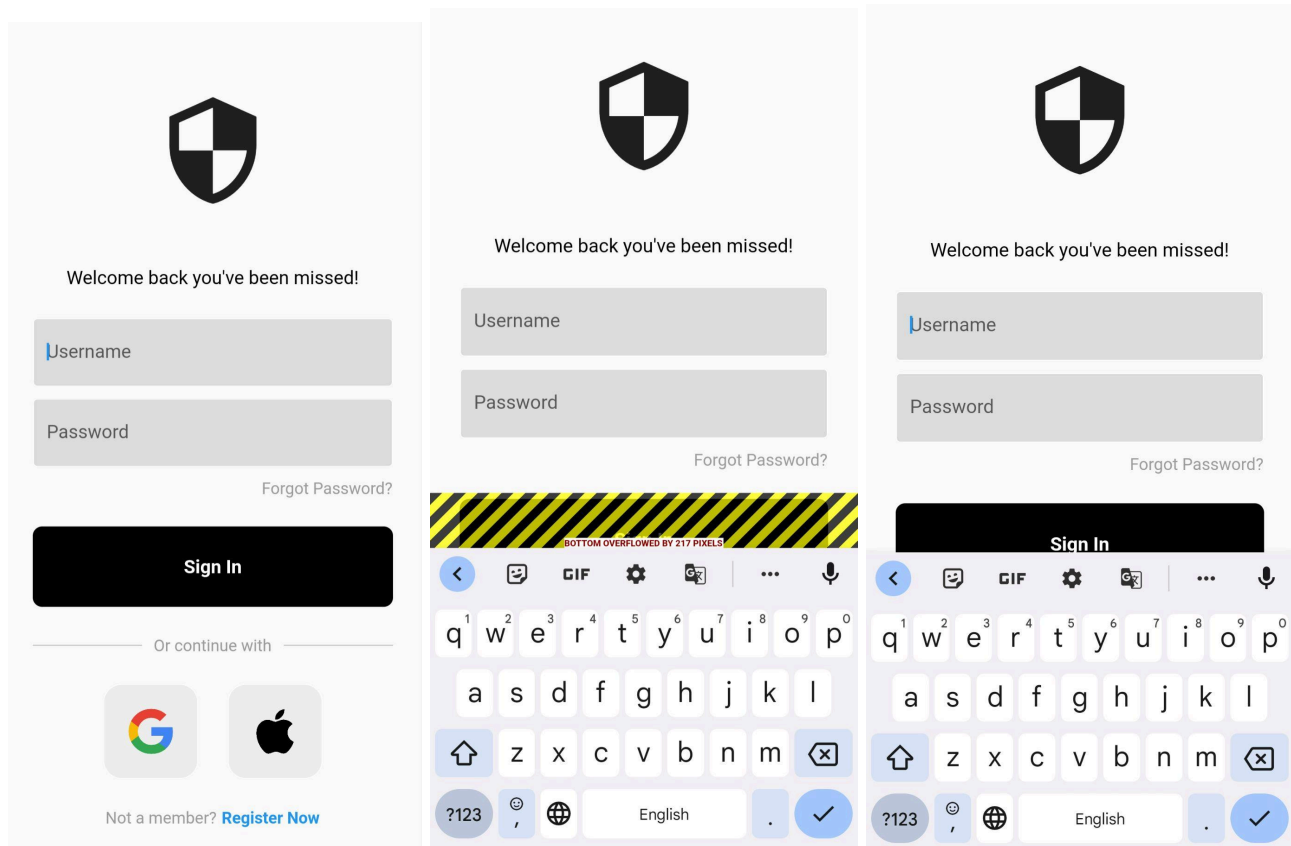
4. Set Goals Page(2)



Error: Overflow of text Widget

Solution: Added SingleChildScrollView Class in the body section.

5. Login Page



Testing Date - 29th January, 2023

Error - When we click on a TextField then an overflow Error appears in the UI.

Solution - Wrap the Column with the SingleChildScrollView() Widget

Code Snippet -

```

96 | @override
97 | Widget build(BuildContext context) {
98 |   return Scaffold(
99 |     backgroundColor: Colors.grey[10],
100 |     body: SafeArea(
101 |       //we have used SafeArea - widget here because it avoids the area in the notch area
102 |       child: Center(
103 |         //used Center widget to align everything in center of the screen
104 |         child: SingleChildScrollView(
105 |           child: Column(
106 |             mainAxisAlignment: MainAxisAlignment.center,
107 |             // it was here the error where I wrote const keyboard after children which c

```

6. Circular Progress Indicator

View the video which we have recorded – with error and without error

Link - <https://drive.google.com/drive/folders/1z5ofn60dhdvV9S4iTWd7KMVN3qSqBYAk?usp=sharing>

Testing Date - 6th February, 2023

Error - The problem we had faced is that when you use the circular page indicator, the indicator keeps on rotating and never stops when user clicks on Sign In.

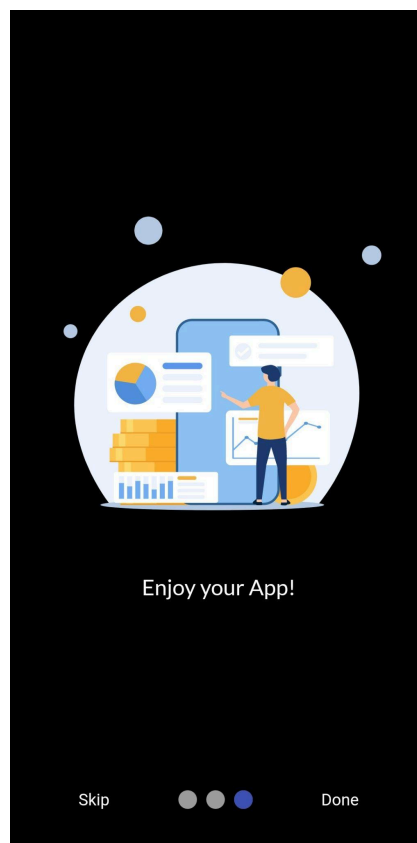
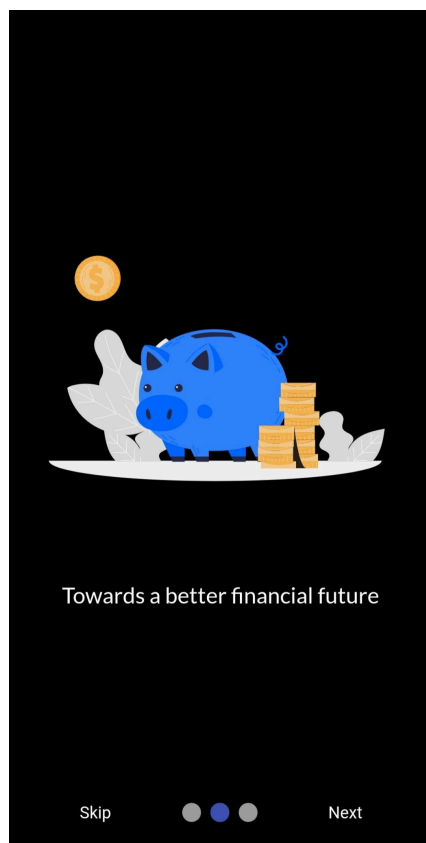
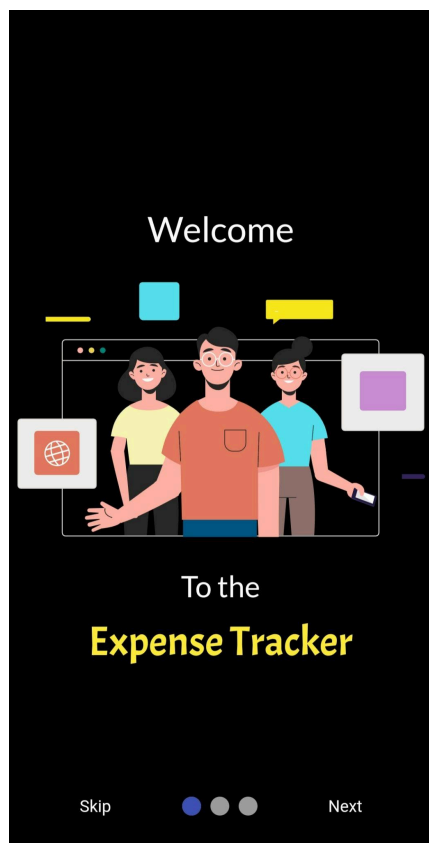
Solution – add `Navigator.pop(context);` inside the `signUserind()` method

Code Snippet -

```
50      //
51    } on FirebaseAuthException catch (e) {
52      //pop the loading circle
53      Navigator.pop(this.context);
54      //
55      //show error message
56      showErrorMessage(e.code);
57    }
58  }
59  //sign user in method-----
60
```

6. UI Screenshots

Here is the User Interface of our Application.





Hello Again

Welcome back you've been missed!

[Forgot Password?](#)

Sign In

Or continue with




Not a member? [Register Now](#)



Enter your Email and we will send you a password reset link

Reset Password

Sign Up

Already have an account? [Login Now](#)



Let's create an account for you

Expense Tracker

Hello, Sundar



Total Balance

₹ 2,957

Income

₹ 1,450

Expenses

₹ 450

Expense Categories





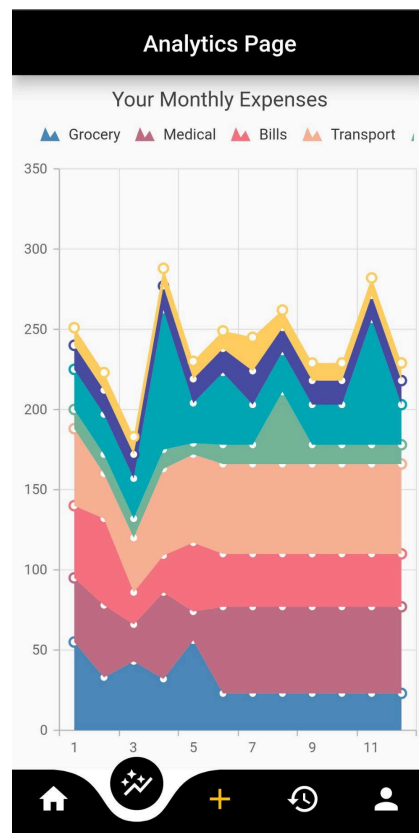










Add Transaction

1 Step 1

Income ☐ Expense ☒

Medical

CONTINUE CANCEL

2 Step 2

3 Step 3

4 Complete

ADD TRANSACTION

Recent Transactions

	Jurassic Park Movie 2023-04-19 05:02:00.000	- ₹ 500.00
	Rewards Point 2023-03-17 20:12:00.000	+ ₹ 100.00
	Buy Back 2023-04-14 02:01:00.000	+ ₹ 50.00
	MacBook Air 2023-03-25 10:20:00.000	- ₹ 24000.00
	Apolo hospital 2023-03-20 08:59:00.000	- ₹ 12444.00
	Dominos 2023-04-23 04:45:00.000	- ₹ 500.00
	Zara 2023-02-28 23:34:00.000	+ ₹ 1258.00
	Hello 2023-02-23 10:47:49.365	- ₹ 789.00

User Page

Hi, sundar@google.com

My Profile →

Change Password →

Set Goal →

Logout

My Profile

Sundar

Pichai

sundar@google.com

738291333333

23

Male

7281919

My Profile

Sundar

Pichai

sundar@google.com

738291333333

23

Male

7281919

UPDATE

Set Goal

Grocery	₹ 8739
Entertainment	₹ 6486
Bills	₹ 6802
Transportation	₹ 4865
Food	₹ 3559
Medical	₹ 4955
Shopping	₹ 2883
Miscellaneous	₹ 4550

7. Databases

The below images shows the data that is stored in Firebase Authentication.

Search by email address, phone number, or user UID Add user				
Identifier	Providers	Created ↓	Signed In	User UID
rhea.yadav@xaviers.edu.in		Feb 28, 2023	Feb 28, 2023	vipk6jbhsbUzyPp9DEwlpP1qnW33
satya@microsoft.com		Feb 27, 2023	Feb 28, 2023	A2LjEeHqicXVfOC3Q0Lj1MRoS1
h1@gmail.com		Feb 27, 2023	Feb 27, 2023	dKJrqo0EGXdrpg0GIANMNoc7V5i2
sundar@google.com		Feb 26, 2023	Mar 1, 2023	tFevtlal8KRQf0pfe91tS7PY5VT2
test9@gmail.com		Feb 26, 2023	Feb 26, 2023	6E2A9HxfPwSx7hPHrNOb55V8fIV2
naman.jain@xaviers.edu.in		Feb 21, 2023	Mar 1, 2023	kKQDmFb9YsdOe8gbsJW9C7DhD...
test8@gmail.com		Feb 21, 2023	Feb 27, 2023	z2j9812dIYTHcD15DJI1hec7IXz1
test7@gmail.com		Feb 21, 2023	Feb 21, 2023	hBwJNTs6qQRuh5Gj6fw1dBz7cOs2
test6@gmail.com		Feb 21, 2023	Feb 21, 2023	wSTPlzwnmFZKcl7tYZmqzoKgKz...
test@jdns.com		Feb 21, 2023	Feb 21, 2023	Fr3ZpkrlaKeGBSSxJKw3R7VQVLz1
test2@gmail.com		Feb 21, 2023	Feb 21, 2023	PVReeKQtslaJKEtoLydMKNCOA103
alpha8@gmail.com		Feb 18, 2023	Feb 18, 2023	PpiPeAaSVbPi73ph5xVAqrQahLz1
alpha7@gmail.com		Feb 18, 2023	Feb 18, 2023	luR7GJPFR1ePWHrsfmjrCRge2T2
alpha5@gmail.com		Feb 18, 2023	Feb 18, 2023	rBYKoiGdK1fFoZ0dzzLswHMz7Ug2
alpha3@gmail.com		Feb 18, 2023	Feb 18, 2023	25xrGs0zTeP2Brp6WghzvMCGOG...
alpha2@gmail.com		Feb 7, 2023	Feb 7, 2023	9mWHgwHDqbbVKnlIYk7kSyK6U...
test@gmail.com		Feb 6, 2023	Feb 26, 2023	RhspSrXdErS2StNfnQ6Ed8I4A02
Rows per page: 100 1 - 18 of 18				

Below is the **Usage Statistics** of the Firebase Authentication.

Daily Active Users

Current billing period
Feb 1 - Mar 1 (PST)

Daily active users

18 peak



Monthly Active Users

Current billing period
Feb 1 - Mar 1 (PST)

Monthly active users

37



Below are the multiple ways we had tested to structure our database and finally we have concluded to one type of structuring the database.

Panel view Query builder

login-and-signin-9631d userDataCollection SS7GhvmAZNukBFBMpUI1

+ Start collection

Naman Jain

sliders

transactions

userDataCollection >

+ Add document

SS7GhvmAZNukBFBMpUI1 >

ZIHnzy2y0ak5ovAG5DKA

fz1XZhgKCQLfDDYUpfPB

h1@gmail.com

hWJs8EXrJzjVdwnPg8y4

naman.jain@xaviers.edu.in

rhea.yadav@xaviers.edu.in

satya@microsoft.com

sundar@google.com

+ Start collection

+ Add field

Age: 20

Email: "test6@gmail.com"

First Name: "Test6"

Gender: "Male"

Last Name: "Tester"

Phone number: 773737

Total Amount: 2000

Database location: eur3

Panel view Query builder

login-and-signin-9631d userDataCollection naman.jain@xav.

+ Start collection

Naman Jain

sliders

transactions

userDataCollection >

+ Add document

SS7GhvmAZNukBFBMpUI1

ZIHnzy2y0ak5ovAG5DKA

fz1XZhgKCQLfDDYUpfPB

h1@gmail.com

hWJs8EXrJzjVdwnPg8y4

naman.jain@xaviers.edu.in >

rhea.yadav@xaviers.edu.in

satya@microsoft.com

sundar@google.com

+ Start collection

+ Add field

Email: "naman.jain@xaviers.edu.in"

First Name: "Naman"

Food: {Dabeli: {Date & Time: Feb...}}

Gender: "Male"

Grocery: {Bread: {Note: "", Date & ...}}

Last Name: "Jain"

Medical: {Hospital Charges: {Date & ...}}

Phone Number: "8282828"

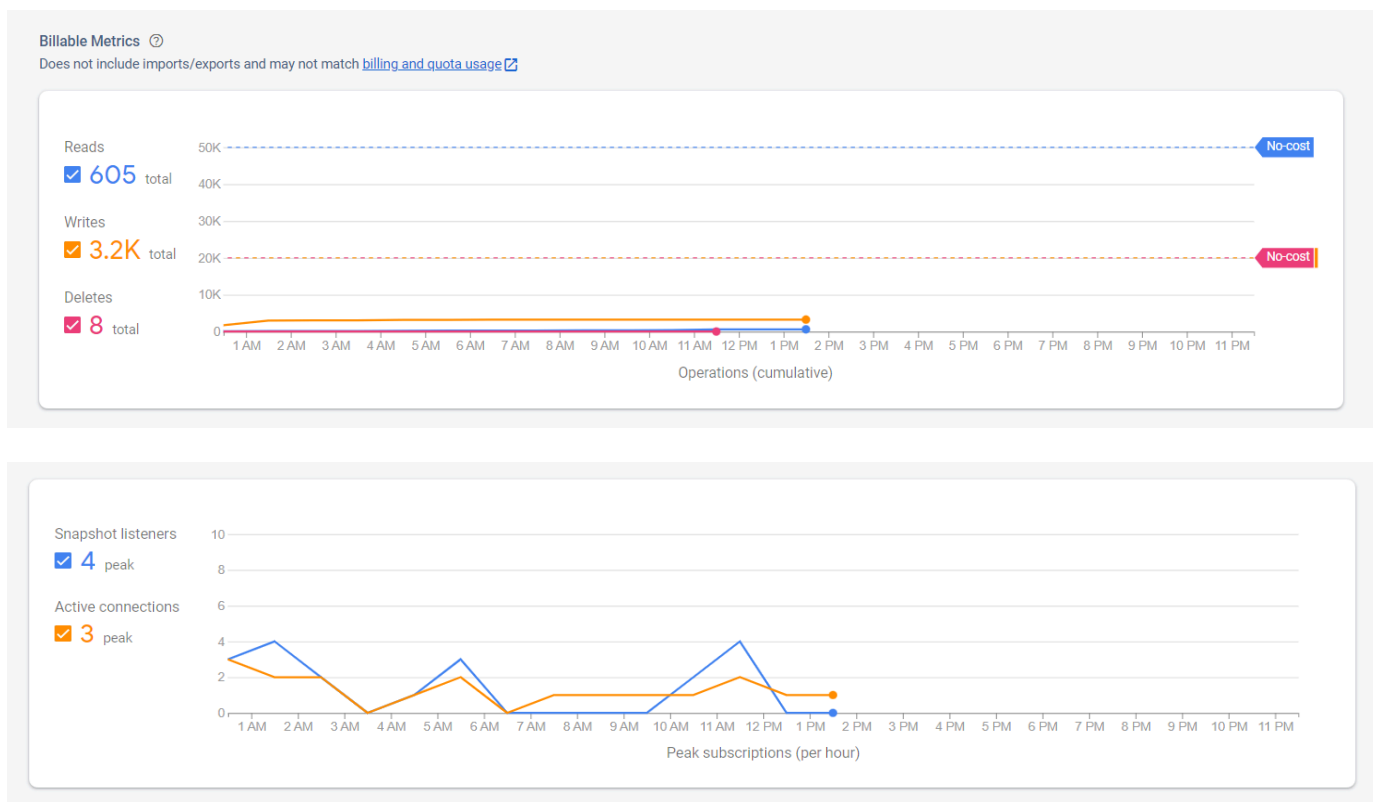
Profile "https://firebasestorage.googleapis.com/v0/b/login-and-signin-9631d.appspot.com/o/profile%2FkKQDmFb9YsdOe8gbsJW9C7DhD9j17alt=media&token=d46d69c2-ac12-4be9-a78b-ae79947ac288"

Total Amount: "82828"

Transportation: {College Bus: {Note: "", D...}}

Database location: eur3

Below is the **Usage Statistics** of the Firebase Firestore



Below is how we used Firebase Storage to store users' profile images.

gs://login-and-signin-9631d.appspot.com > profile

☐	Name	Size	Type	Last modified
☐	6E2A9HxfPwSx7hPHrNob55V8fIV232	82.86 KB	image/jpeg	Feb 26, 2023
☐	A2LjEeHqiqXVfOC3Q0Lj1MRoS120	37.62 KB	image/jpeg	Feb 27, 2023
☐	A2LjEeHqiqXVfOC3Q0Lj1MRoS145	37.62 KB	image/jpeg	Feb 27, 2023
☐	A2LjEeHqiqXVfOC3Q0Lj1MRoS153	37.62 KB	image/jpeg	Feb 27, 2023
☐	A2LjEeHqiqXVfOC3Q0Lj1MRoS16	37.62 KB	image/jpeg	Feb 27, 2023
☐	A2LjEeHqiqXVfOC3Q0Lj1MRoS17	37.62 KB	image/jpeg	Feb 27, 2023
☐	QeTYnhHRP2gMI6Zk3snz5u7WlvG214	22.38 KB	image/jpeg	Feb 27, 2023
☐	W0cq4ofGovVXiKNpLUkinreoWne253	22.38 KB	image/jpeg	Feb 27, 2023
☐	dKJrq0EGXdrpg0GIANMNoc7V5i212	22.38 KB	image/jpeg	Feb 27, 2023

Upload file + ⋮

A2LjEeHqiqXVfOC... ×

Name
[A2LjEeHqiqXVfOC3Q0Lj1MRoS153](#) ⓘ

Size
38,523 bytes

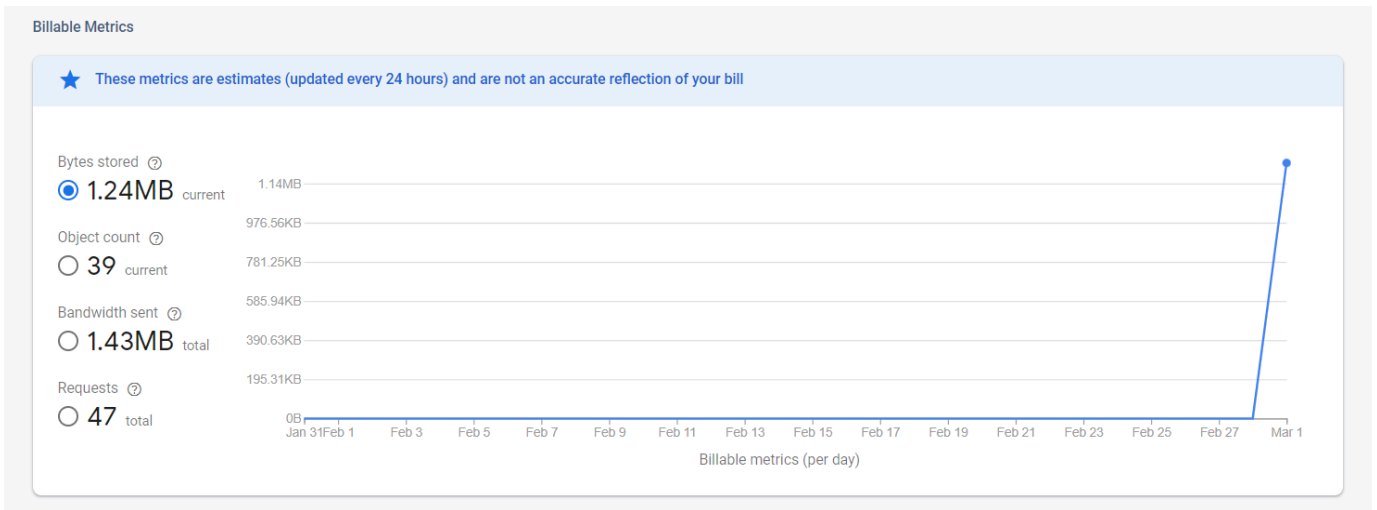
Type
image/jpeg

Created
Feb 27, 2023, 1:57:56 AM

Updated
Feb 27, 2023, 1:57:56 AM

File location ▼

Below is the Usage statistics of the Firebase Storage



8. Testing Used

We have tested our Application on Firebase's testing lab and also done Unit Testing. We have successfully tested our Application on mobile devices like Realme 9 5G and Moto G52 and on Android 13 and Android 12.

We have used Firebase's Test Lab Platform to test our application where we have used **Robo Type Test**.

Robo Test - Firebase Test lab

Robo Type Test is a type of automated testing available in Firebase Test Lab for Android. It simulates user interactions by sending random keystrokes to the app, including letters, numbers, and special characters. The purpose of this test is to identify issues related to user input, such as crashes or unexpected behavior when typing.

During the test, the Robo agent explores the app and generates a sequence of events that simulate user actions. It observes the app's response to each input and tries to infer the app's behavior to improve the test coverage. Robo Type Test is especially useful for apps that rely heavily on user input, such as messaging or email apps.

After the test is complete, Firebase Test Lab provides a detailed report with logs and screenshots of the app's behavior during the test. This report can help developers identify and fix issues related to user input before releasing the app to the public.

Here are the details the Robo Test has generated

Test Lab > login_and_signup > matrix-2acceb78pltp8

Robo test, Pixel 5, API Level 30

✓ Passed 3/1/23, 2:56 AM 4m 40s Portrait English (United States)

[View test artifacts](#)

[Test Issues](#) Robo Logs Screenshots Videos Performance Accessibility

! Usage of non-SDK interfaces

! **Non-SDK API Usage Violation** [Learn more about non-SDK interface restrictions](#)

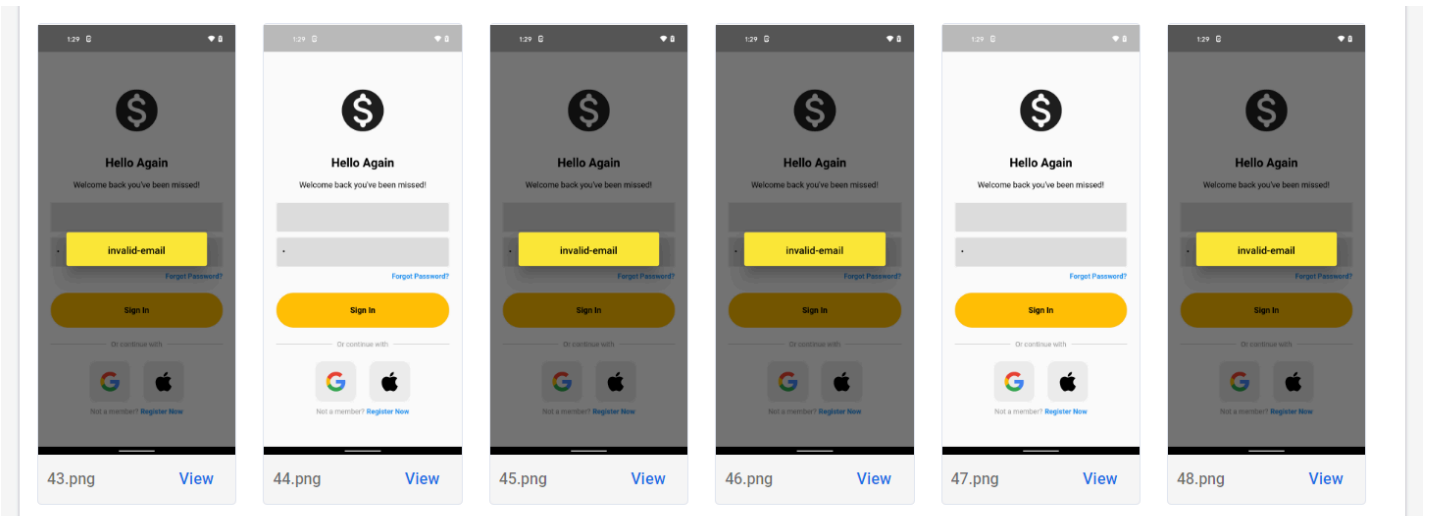
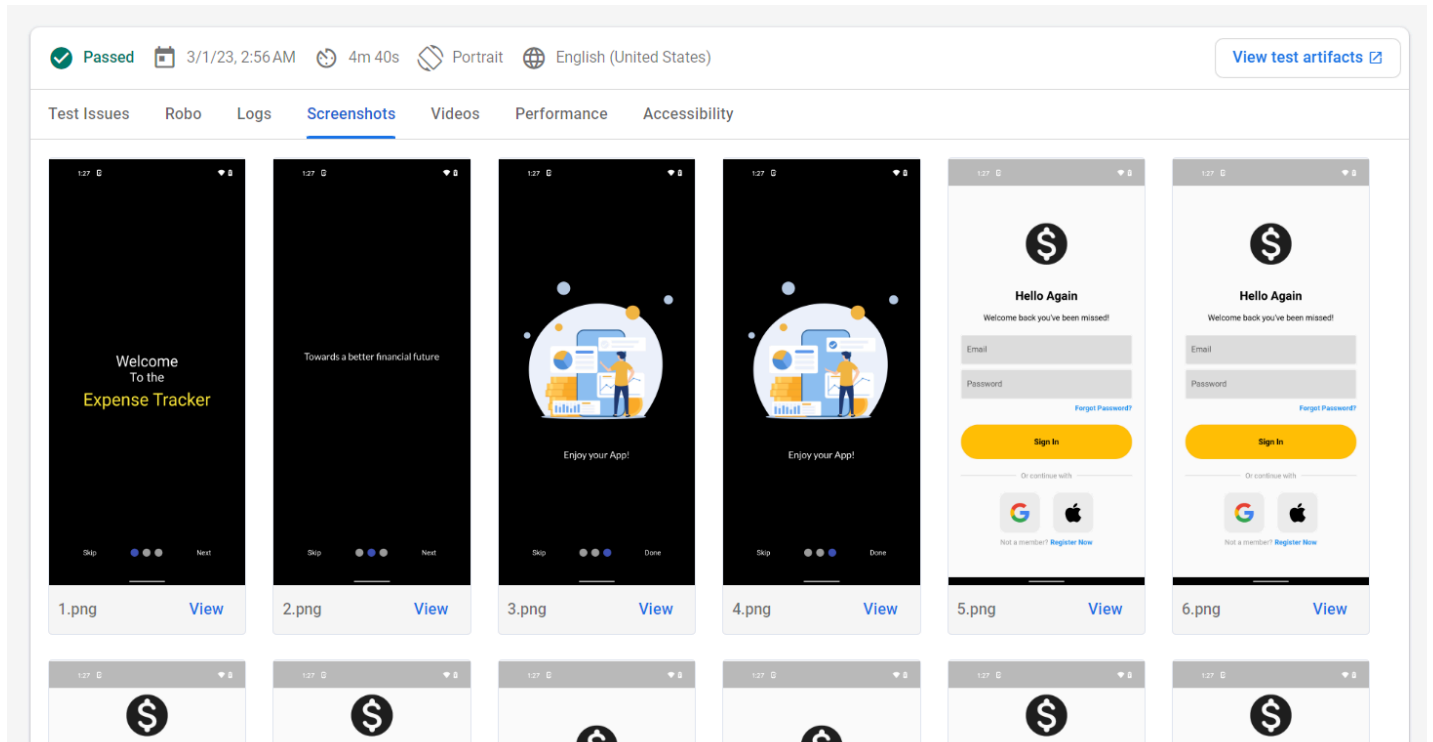
Your app uses 4 non-SDK interfaces, which are incompatible with Android P+.

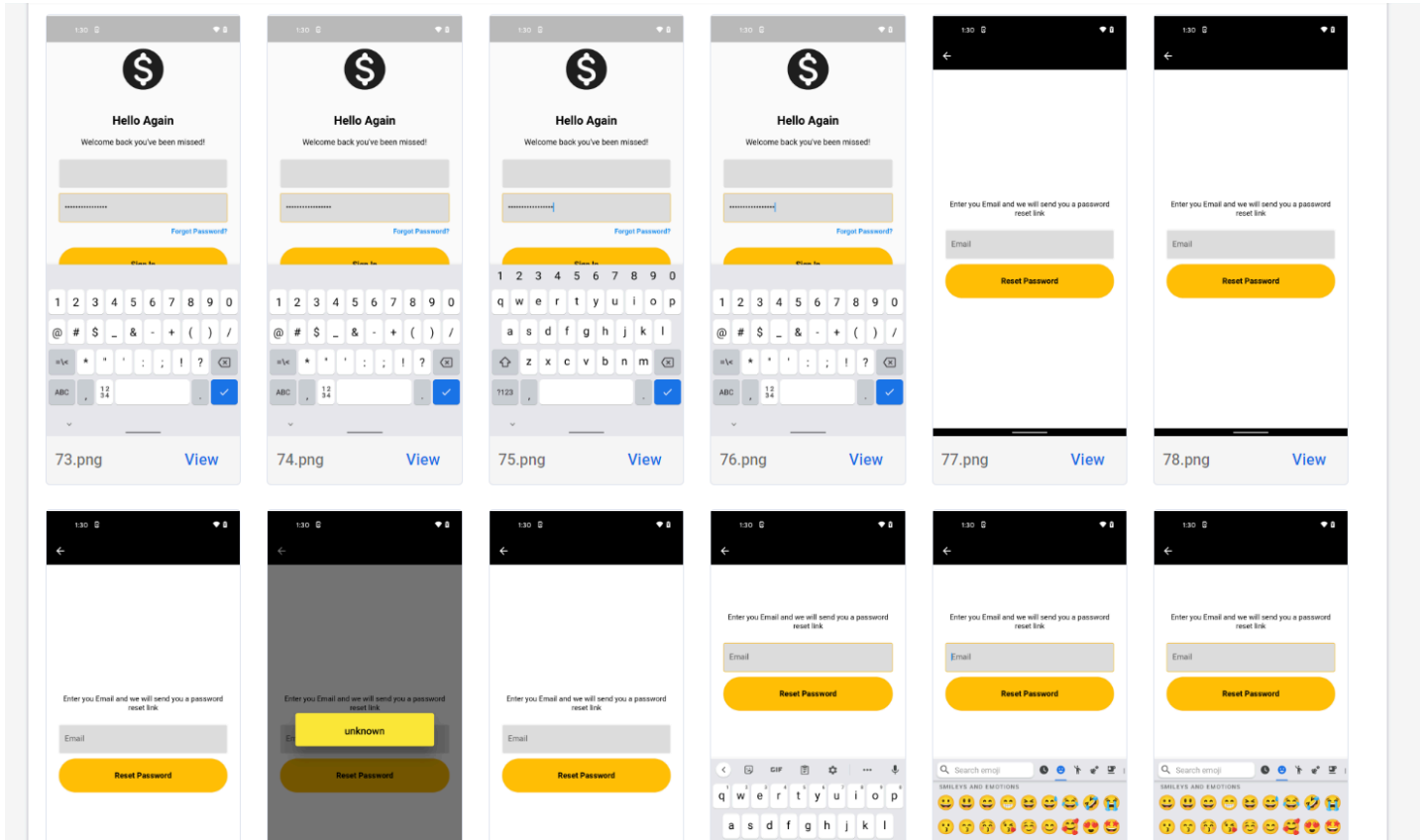
```
Landroid/view/accessibility/AccessibilityNodeInfo;->mChildNodeIds:Landroid/util/LongArray;
```

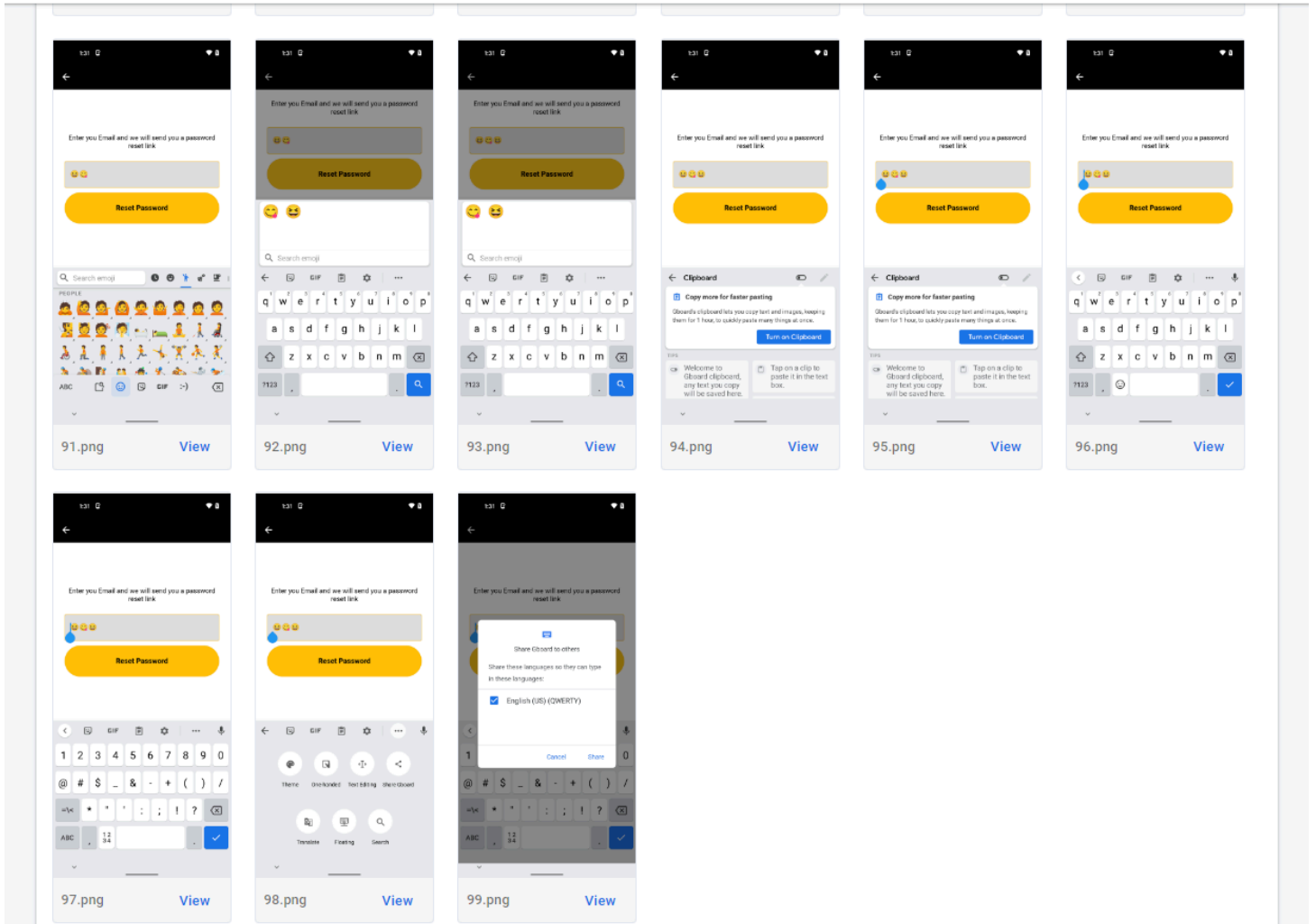
```
Landroid/view/accessibility/AccessibilityRecord;->getSourceNodeId()J
```

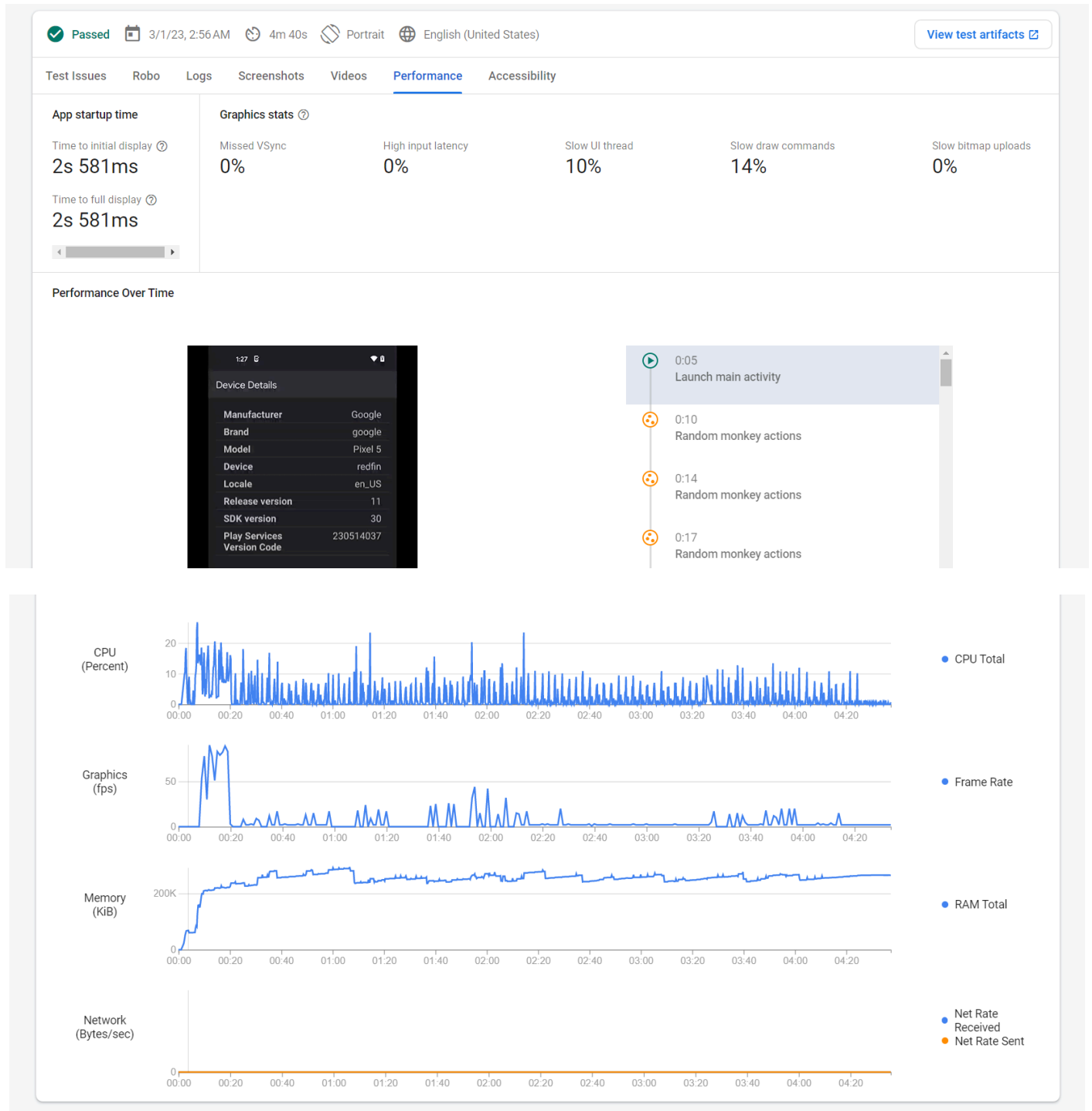
```
Landroid/util/LongArray;->get(I)J
```

```
Landroid/view/accessibility/AccessibilityNodeInfo;->getSourceNodeId()J
```







Passed

3/1/23, 2:56 AM

4m 40s

Portrait

English (United States)

View test artifacts

Test IssuesRoboLogsScreenshotsVideosPerformanceAccessibility

★ Accessibility issues found by automated testing are displayed here, along with associated recommendations.

Learn more

Overview

1 issue found

Issue types	Warnings	Minor issues	Tips
Touch target size	0	0	0
Low contrast	0	0	0
Content labeling	1	0	0
Implementation	0	0	0

Touch target size (0)

No issues found.

Low contrast (0)

No issues found.

Content labeling (1)

1:31

←

Enter your Email and we will send you a password reset link.

Email

Reset Password

Search emoji

SMILEYS AND EMOJIONS

ABC

98 examples

View

UNIT Test's

We have tested different areas of our application and made videos of our testing. So that people can understand how we did our test. You can access our videos from the below link.

Video Link - <https://drive.google.com/drive/folders/1z5ofn60dhdvV9S4iTWd7KMVN3qSqBYAk?usp=sharing>

Testing Date - 18th February, 2023

Testing 1- We have tested the forgot password page the reason we are testing this page because we wanted to see when a user click on forgot password the forgot password page should appear and when the user enters their email address and click on reset password button then the Firebase Authentication should send them the email to reset the password and also a pop up message should be sent to them notifying that the password reset link has been send to them.

Testing Date - 18th February, 2023

Testing 2- We also tested the pop up messages for example if a user whose email address is not registered in our Firebase then our pop up message is showing the error message also if a email syntax is wrong user want to reset we have also on proper message which is called invalid email which is working the main motive of this part is just to do a normal testing.

Testing Date - 14th February, 2023

Testing 2- We tested whether our application is able to execute the login and signout feature or not.

9. Learnings & Challenges

- From working with a slider in Flutter, we learned how to create a user interface that allows users to interact with my application through input devices. I learned how to create custom widgets to handle user input and how to work with callbacks to update the user interface in response to changes in user input.
- Working with forms in Flutter taught us how to collect user input and validate that input. We learned how to use different input types, such as text fields, radio buttons, and drop-down menus, to collect different types of input. We also learned how to use form controllers to manage user input and how to use form validation to ensure that user input meets certain criteria.
- Working with Firestore in Flutter taught us how to integrate our application with a cloud-based database service. We learned how to use the Firebase SDK to create, read, update, and delete data in the database. We also learned how to use Firebase Authentication to secure access to my database and how to use Firebase Cloud Functions to perform server-side operations. Additionally, We learned how to use Firebase Cloud Messaging to send push notifications to my users.
- Learned how to design an interactive UI for our users using different Widgets of Flutter.
- We also learned if the icons touch or go to the top of any mobile, so to rectify it and keep it within our application, we have to use the SafeArea() Widget.
- How to create interactive graphs.
- We also explored various packages of Flutter and learnt how to integrate it in our app.
- We have also maintained the code reusability across our app.

For Example - there are in total 19 TextFields in our app, and we have not written 19 times the same code. Instead we have made one class and call that class wherever we needed. And we have done that same techniques with buttons too.

- How to navigate between the pages.

10. Future Scope

- Integration with financial institutions: Many users would find it useful to be able to link their bank accounts, credit cards, and other financial accounts to their expense tracker app. This would allow for automatic tracking of expenses and more accurate budgeting.
- Machine learning-based spending analysis: By using machine learning algorithms, the app could analyze spending patterns and provide personalized insights and recommendations for optimizing spending habits.

- Multi-currency support: For users who travel frequently or have international expenses, support for multiple currencies would be helpful. The app could automatically convert expenses to the user's home currency for easier tracking.
- Expense sharing and splitting: For users who share expenses with friends or family, the app could provide features for splitting bills and tracking shared expenses.
- Integration with digital wallets and payment apps: As digital payments become more popular, integrating with popular digital wallets and payment apps such as PayPal and Venmo could be useful for users.
- Overall, there are many potential features that could be added to an expense tracker app to make it more useful and appealing to users. By listening to user feedback and keeping up with industry trends, developers can continue to improve and expand the functionality of the app.

11. References

- Official Flutter documentation: <https://flutter.dev/docs>
- Flutter packages and plugins from pub.dev: <https://pub.dev/>
- Stack Overflow community forums: <https://stackoverflow.com/>
- YouTube tutorials from reputable channels:
 - ▶ flutter finance app - full course - Flutter UI Design
 - ▶ Flutter Firebase Firestore CRUD Operations | Create, Read, Update & Delete

Flutter-Widget of the week, Flutter-Package of the Week SERIES at <https://www.youtube.com/@flutterdev>
- Firestore documentation: <https://firebase.google.com/docs/firestore>
- Practical sessions in the Lab and course materials on Google Classroom
- Firebase CLI Documentation - <https://firebase.google.com/docs/cli>
- Here are the references of the packages we have used : -
 - cupertino_icons: ^1.0.2 - https://pub.dev/packages/cupertino_icons
 - firebase_auth: ^4.2.5 - https://pub.dev/packages/firebase_auth
 - firebase_core: ^2.4.1 - https://pub.dev/packages/firebase_core
 - cloud_firestore: ^4.3.1 - https://pub.dev/packages/cloud_firestore
 - smooth_page_indicator: ^1.0.0+2 - https://pub.dev/packages/smooth_page_indicator
 - lottie: ^2.2.0 - <https://pub.dev/packages/lottie>
 - google_fonts: ^4.0.3 - https://pub.dev/packages/google_fonts
 - curved_navigation_bar: ^1.0.3 - https://pub.dev/packages/curved_navigation_bar
 - image_picker: ^0.8.6+3 - https://pub.dev/packages/image_picker
 - firebase_storage: ^11.0.14 - https://pub.dev/packages/firebase_storage
 - shared_preferences: ^2.0.18 - https://pub.dev/packages/shared_preferences
 - confirm_dialog: ^1.0.1 - https://pub.dev/packages/confirm_dialog
 - syncfusion_flutter_charts: ^20.4.51 - https://pub.dev/packages/syncfusion_flutter_charts
 - date_field: ^3.0.2 - https://pub.dev/packages/date_field